Intermediate modular quotients with gained good reduction

Let N be squarefree and put $W = W_N$. The table classifies every positive-genus quotient $X_0(N)/H$ with $0 < H < W$ for which the Jacobian gains good reduction at a prime dividing N. It uses the complete lacking-sign data in Tables 2–5 of Padurariu–Park–Voight.

For $M \mid N$, the level-M factor with Atkin–Lehner character ϵ is retained exactly when ϵ is trivial on $H \cap W_M$. A prime $p \mid N$ belongs to p_J precisely when every retained factor with $p \mid M$ is lacking.

There are 73 quotients at 41 levels. The displayed generators are the canonical reduced-row-echelon basis relative to the increasing primes dividing N.

Table 1: Positive-genus intermediate modular quotients with a gained prime.

N	H	p_J	N	H	p_J
22	$\langle w_2 \rangle$	$\{2\}$	86	$\langle w_{86} \rangle$	$\{2\}$
22	$\langle w_{22} \rangle$	$\{2\}$	87	$\langle w_{87} \rangle$	$\{3\}$
30	$\langle w_3 \rangle$	$\{2\}$	94	$\langle w_{94} \rangle$	$\{2\}$
30	$\langle w_6 \rangle$	$\{2\}$	95	$\langle w_{95} \rangle$	$\{5\}$
30	$\langle w_{30} \rangle$	$\{2\}$	105	$\langle w_3, w_{35} \rangle$	$\{7\}$
30	$\langle w_2, w_3 \rangle$	$\{2\}$	105	$\langle w_5, w_7 \rangle$	$\{5\}$
30	$\langle w_{10}, w_3 \rangle$	$\{2\}$	105	$\langle w_{21}, w_5 \rangle$	$\{3\}$
33	$\langle w_{33} \rangle$	$\{3\}$	110	$\langle w_{110} \rangle$	$\{2\}$
34	$\langle w_2 \rangle$	$\{2\}$	110	$\langle w_2, w_{55} \rangle$	$\{2, 5\}$
34	$\langle w_{34} \rangle$	$\{2\}$	110	$\langle w_{10}, w_{11} \rangle$	$\{2\}$
38	$\langle w_{38} \rangle$	$\{2\}$	110	$\langle w_{22}, w_5 \rangle$	$\{2\}$
42	$\langle w_{14} \rangle$	$\{2\}$	111	$\langle w_{111} \rangle$	$\{3\}$
42	$\langle w_2, w_3 \rangle$	$\{3\}$	114	$\langle w_2, w_{19} \rangle$	$\{2\}$
42	$\langle w_2, w_7 \rangle$	$\{2\}$	114	$\langle w_{38}, w_3 \rangle$	$\{2\}$
42	$\langle w_2, w_{21} \rangle$	$\{3\}$	119	$\langle w_{119} \rangle$	$\{7\}$
42	$\langle w_6, w_7 \rangle$	$\{2\}$	130	$\langle w_2, w_{13} \rangle$	$\{2\}$
46	$\langle w_{46} \rangle$	$\{2\}$	130	$\langle w_{26}, w_5 \rangle$	$\{2\}$
51	$\langle w_{51} \rangle$	$\{3\}$	134	$\langle w_{134} \rangle$	$\{2\}$
55	$\langle w_{55} \rangle$	$\{5\}$	146	$\langle w_{146} \rangle$	$\{2\}$
62	$\langle w_{62} \rangle$	$\{2\}$	159	$\langle w_{159} \rangle$	$\{3\}$
66	$\langle w_{66} \rangle$	$\{2\}$	170	$\langle w_{34}, w_5 \rangle$	$\{2\}$
66	$\langle w_2, w_3 \rangle$	$\{2\}$	174	$\langle w_2, w_{87} \rangle$	$\{3\}$
66	$\langle w_2, w_{33} \rangle$	$\{2, 3\}$	182	$\langle w_{26}, w_7 \rangle$	$\{2\}$
66	$\langle w_{22}, w_3 \rangle$	$\{2\}$	186	$\langle w_{62}, w_3 \rangle$	$\{2\}$
69	$\langle w_{69} \rangle$	$\{3\}$	194	$\langle w_{194} \rangle$	$\{2\}$
70	$\langle w_{14} \rangle$	$\{2\}$	195	$\langle w_{39}, w_5 \rangle$	$\{3\}$
70	$\langle w_2, w_7 \rangle$	$\{2\}$	206	$\langle w_{206} \rangle$	$\{2\}$
70	$\langle w_2, w_{35} \rangle$	$\{5\}$	222	$\langle w_2, w_{111} \rangle$	$\{3\}$
70	$\langle w_{10}, w_7 \rangle$	$\{2\}$	222	$\langle w_{74}, w_3 \rangle$	$\{2\}$
70	$\langle w_{14}, w_5 \rangle$	$\{2\}$	230	$\langle w_{10}, w_{23} \rangle$	$\{2\}$
74	$\langle w_{74} \rangle$	$\{2\}$	230	$\langle w_{46}, w_5 \rangle$	$\{2\}$
78	$\langle w_{26} \rangle$	$\{2\}$	231	$\langle w_{21}, w_{11} \rangle$	$\{3\}$
78	$\langle w_{39} \rangle$	$\{3\}$	255	$\langle w_{51}, w_5 \rangle$	$\{3\}$
78	$\langle w_2, w_{13} \rangle$	$\{2\}$	266	$\langle w_{38}, w_7 \rangle$	$\{2\}$
78	$\langle w_2, w_{39} \rangle$	$\{3\}$	330	$\langle w_2, w_{33}, w_{55} \rangle$	$\{2\}$
78	$\langle w_3, w_{13} \rangle$	$\{3\}$	330	$\langle w_{22}, w_3, w_5 \rangle$	$\{2\}$
78	$\langle w_{26}, w_3 \rangle$	$\{2\}$			

Source: O. Padurariu, S. W. Park, and J. Voight, *Unexpected primes of good reduction in quotients of modular and Shimura curves*, `https://arxiv.org/abs/2607.11248`.